

# Lab 04 — Equivalent Networks and Superposition

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## Description

This lab covers three circuits. The first (Figure 4.4) is a single-source resistor ladder, where the goal is to find its Thévenin equivalent and check it against a direct measurement of the load voltage. The second (Figure 4.5) is a linear circuit with two sources, used to test the superposition principle. The third (Figure 4.6) adds a 1N4148 diode in series with the load to see whether superposition still works once a nonlinear part is in the circuit. Each circuit was simulated in LTspice 26 with a DC operating point and then built on the trainer and measured with a voltmeter.

## Procedure

1. Enter Figures 4.4, 4.5, and 4.6 as LTspice schematics and run a DC operating-point (.op) simulation of each.
2. For Figure 4.5, compute  $V_L$  across the  $1\text{ k}\Omega$  load by superposition, then reproduce it in simulation with both sources present and with each source replaced by a short (0 V source).
3. *Task 1:* build Figure 4.4; measure  $V_L$  for two load resistors other than  $2.2\text{ k}\Omega$  ( $4.7\text{ k}\Omega$  and  $1\text{ k}\Omega$ ), then for  $R_L = 2.2\text{ k}\Omega$ ; back out the Thévenin equivalent and predict the  $2.2\text{ k}\Omega$  reading.
4. *Task 2:* build Figure 4.5; measure  $V_L$  with both sources, then  $V_{L,1}$  and  $V_{L,2}$  with each source individually shorted.
5. *Task 3:* repeat Task 2 on Figure 4.6 (series diode added).

## Calculations

**Superposition, Figure 4.5.** Take the bottom rail as ground. The 3 V source fixes node  $A = 3\text{ V}$ ; the 5 V source (its + terminal at ground) fixes node  $B = -5\text{ V}$ . The load  $R_L = 1\text{ k}\Omega$  joins mid-nodes  $P$  and  $Q$ , where  $R_2(2.2\text{ k}\Omega)$  ties  $Q$  to  $A$ ,  $R_3(4.7\text{ k}\Omega)$  ties  $Q$  to ground,  $R_5(4.7\text{ k}\Omega)$  ties  $P$  to  $B$ , and  $R_4(2.2\text{ k}\Omega)$  ties  $P$  to ground. Define  $V_L = V(Q) - V(P)$ .

*Source 1 (5 V), 3 V source shorted ( $A = 0$ ):*

$$\frac{Q}{2200} + \frac{Q}{4700} + \frac{Q - P}{1000} = 0, \quad \frac{P + 5}{4700} + \frac{P}{2200} + \frac{P - Q}{1000} = 0 \Rightarrow V_{L,1} = 0.399\text{ V}.$$

*Source 2 (3 V), 5 V source shorted ( $B = 0$ ):*

$$\frac{Q - 3}{2200} + \frac{Q}{4700} + \frac{Q - P}{1000} = 0, \quad \frac{P}{4700} + \frac{P}{2200} + \frac{P - Q}{1000} = 0 \Rightarrow V_{L,2} = 0.511\text{ V}.$$

*Superposition sum:*  $V_{L,1} + V_{L,2} = 0.910\text{ V}$ , matching the direct both-source solve below to within rounding.

**Thévenin extraction, Figure 4.4.** Using  $V_L = V_{th} R_L / (R_L + R_{th})$  with the measured pair  $V_L(4.7\text{ k}\Omega) = 1.299\text{ V}$  and  $V_L(1\text{ k}\Omega) = 0.587\text{ V}$ :

$$R_{th} = \frac{V_L^{(b)} - V_L^{(a)}}{\frac{V_L^{(a)}}{R_L^{(a)}} - \frac{V_L^{(b)}}{R_L^{(b)}}} \approx 2.29\text{ k}\Omega, \quad V_{th} \approx 1.93\text{ V},$$

$$\Rightarrow V_L^{\text{pred}}(2.2\text{ k}\Omega) = 1.93\text{ V} \cdot \frac{2200}{2200 + 2292} = 0.946\text{ V}.$$

```
// Figure 4.5 - nodal analysis, both sources present
// Unknowns: V = [V_Q; V_P]; A = +3 V, B = -5 V
G = [1/2200 + 1/4700 + 1/1000, -1/1000;
     -1/1000, 1/4700 + 1/2200 + 1/1000];
I = [3/2200; -5/4700];

disp("Matrix G:"); disp(G)
disp("Vector I:"); disp(I)

V = G\I; // backslash: numerically stable vs. inv(G)*I
disp("Result V:"); disp(V)

VL = V(1) - V(2); // voltage across the 1 kohm load
disp("V_L=V(1)-V(2):"); disp(VL)
```

Listing 1: SciLab nodal solve, Figure 4.5 (both sources present).

```
"Matrix G:"

  0.0016673  -0.001
-0.001      0.0016673

"Vector I:"

  0.0013636
-0.0010638

"Result V:"

  0.6796776
-0.2304022

"V_L=V(1)-V(2):"

  0.9100798
```

Listing 2: SciLab runtime output.

The SciLab result,  $V_L = 0.910\text{ V}$ , equals the superposition sum  $V_{L,1} + V_{L,2}$ , confirming the hand analysis.

## Simulations

The exported LTspice netlist for Figure 4.4 is shown below; its `.op` run gives  $I(R_L) = 394.7\mu\text{A}$ , i.e.  $V_L(2.2\text{k}\Omega) = 0.868\text{ V}$  for nominal components. The Figure 4.5 operating point reproduces the calculation exactly:  $V_L = 0.910\text{ V}$  (both),  $V_{L,1} = 0.399\text{ V}$ ,  $V_{L,2} = 0.511\text{ V}$ .

```
V1 N001 N004 3
R1 N002 N001 1000
R2 N002 N004 4700
R3 N003 N002 1000
R4 N003 N004 4700
R5 N005 N003 1000
RL N005 N004 2200
.op
.end
```

Listing 3: LTspice netlist, Figure 4.4.

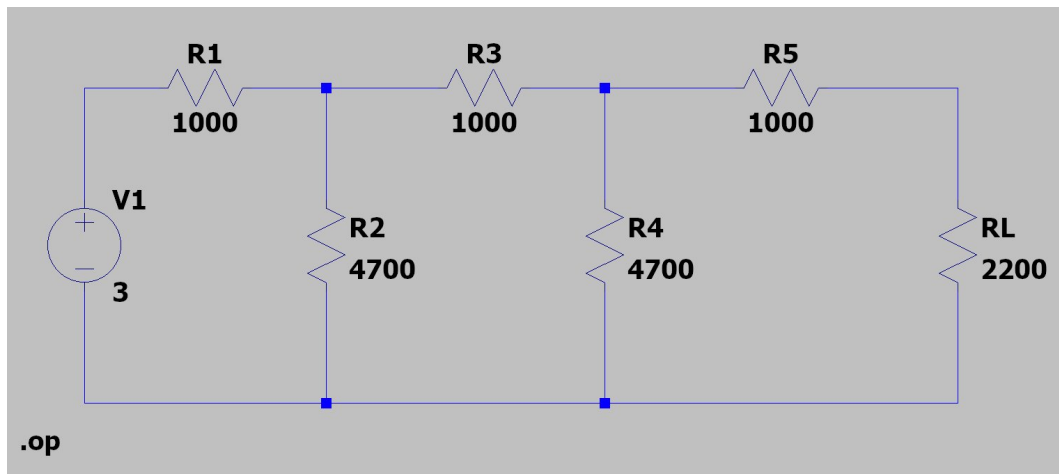


Figure 1: Figure 4.4 schematic (Thévenin ladder), DC operating point.

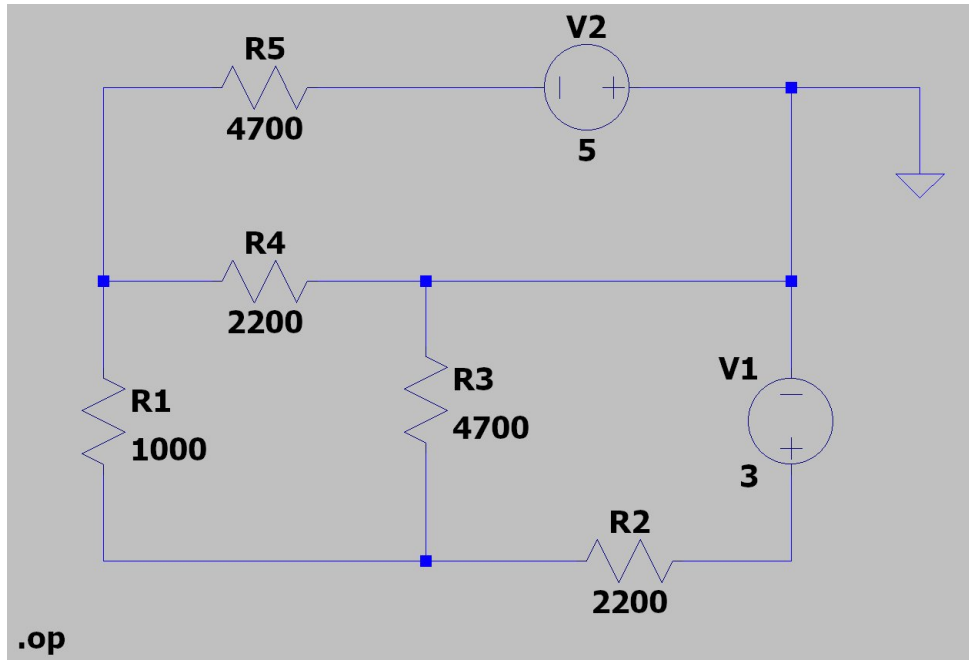


Figure 2: Figure 4.5 schematic (linear, two sources), DC operating point.

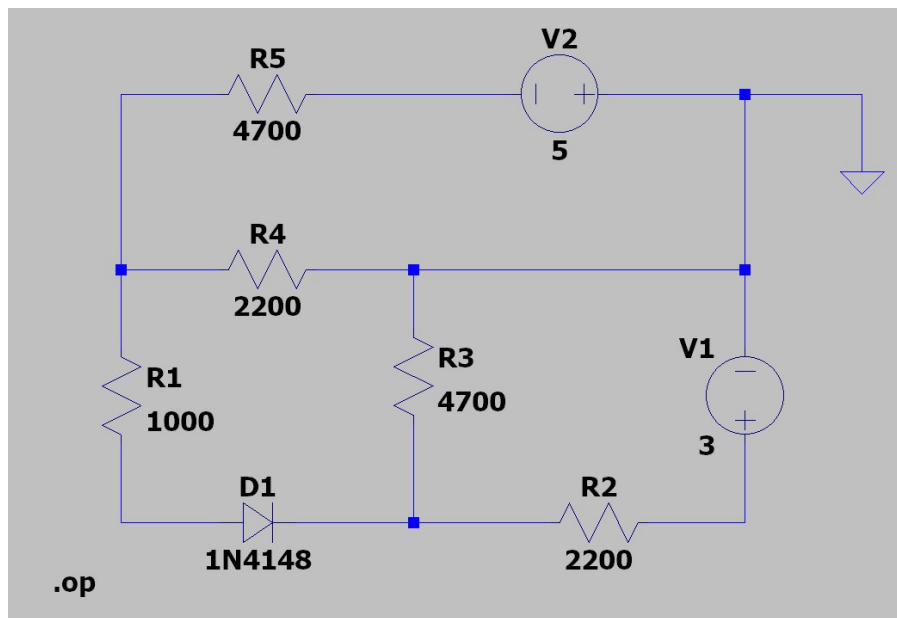


Figure 3: Figure 4.6 schematic (series 1N4148 diode), DC operating point.

The source-removed operating points used to verify superposition on the linear network are shown in Figure 4.

--- Operating Point ---

```
V(n001): -0.996521 voltage
V(n002): -5 voltage
V(n003): -0.597682 voltage
V(n004): 0 voltage
I(R1): -0.00039884 device_current
I(R2): 0.000271673 device_current
I(R3): 0.000127166 device_current
I(R4): 0.000452964 device_current
I(R5): -0.000851804 device_current
I(V1): -0.000271673 device_current
I(V2): -0.000851804 device_current
```

Fig. 4.5, 3 V source shorted

--- Operating Point ---

```
V(n001): 0.766119 voltage
V(n002): 0 voltage
V(n003): 1.27736 voltage
V(n004): 3 voltage
I(R1): -0.00051124 device_current
I(R2): 0.000783019 device_current
I(R3): -0.000271779 device_current
I(R4): -0.000348236 device_current
I(R5): -0.000163004 device_current
I(V1): -0.000783019 device_current
I(V2): -0.000163004 device_current
```

Fig. 4.5, 5 V source shorted

Figure 4: Figure 4.5 (linear) source-removed operating points.

With the diode turned so its anode faces the  $R_3/3$  V-source side, the both-sources run (Figure 5) shows it conducting:  $I(D_1) = I(R_1) = 0.767$  mA, with about a 0.57 V drop across the diode, which leaves 0.767 V across the load. The single-source diode numbers in Tables 1 and 4 are calculated estimates; they match this run to within 0.1%.

--- Operating Point ---

```
V(n001): -0.445067 voltage
V(n002): -5 voltage
V(n003): 0.321765 voltage
V(n004): 0.894342 voltage
V(n005): 3 voltage
I(D1): 0.000766832 device_current
I(R1): -0.000766832 device_current
I(R2): 0.000957117 device_current
I(R3): -0.000190286 device_current
I(R4): 0.000202303 device_current
I(R5): -0.000969135 device_current
I(V1): -0.000957117 device_current
I(V2): -0.000969135 device_current
```

Figure 5: Figure 4.6 both-sources operating point with  $D_1$  conducting:  $I(D_1) = 0.767$  mA,  $V_L = 0.767$  V.

## Pre-Lab Summary (Table 4.1)

Table 1: Table 4.1 — calculated and simulated  $V_L$  across  $R_L$  ( $1\text{ k}\Omega$ ), in volts. The “Sim. w/Diode” both-sources value is from the flipped-diode .op run; the two single-source values are model estimates (within 0.1% of that run).

| $V_L$ (across $R_L/1\text{ k}\Omega$ ) | Calculation | Simulation | Sim. w/Diode |
|----------------------------------------|-------------|------------|--------------|
| $V_1$ & $V_2$ present (A)              | 0.910       | 0.910      | 0.767        |
| $V_1$ only (5 V)                       | 0.399       | 0.399      | 0.268        |
| $V_2$ only (3 V)                       | 0.511       | 0.511      | 0.376        |
| Add lines 2 & 3 (B)                    | 0.910       | 0.910      | 0.644        |
| $\%\Delta$ between A and B             | 0.0%        | 0.0%       | 16.0%        |

Superposition holds for the linear circuit ( $A = B$ ), and fails once the diode is present ( $A \neq B$ ).

## Measurements

Table 2: Task 1 — measured  $V_L$  vs.  $R_L$  (Figure 4.4).

| $R_L$          | Measured $V_L$ (V) |
|----------------|--------------------|
| 1.0 k $\Omega$ | 0.587              |
| 2.2 k $\Omega$ | 0.954              |
| 4.7 k $\Omega$ | 1.299              |

From the  $1\text{ k}\Omega$  and  $4.7\text{ k}\Omega$  readings,  $V_{th} = 1.93\text{ V}$  and  $R_{th} = 2.29\text{ k}\Omega$ , which predict  $V_L(2.2\text{ k}\Omega) = 0.946\text{ V}$  versus the measured  $0.954\text{ V}$  (0.8%).

Table 3: Table 4.2 — Task 2, superposition (linear, Figure 4.5), in volts. %diff is  $|\text{theory} - \text{meas}|/|\text{theory}|$ .

| Parameter           | Calculated | Simulated | Measured | %diff (C v M) | %diff (S v M) |
|---------------------|------------|-----------|----------|---------------|---------------|
| $V_L$               | 0.910      | 0.910     | 0.668    | 26.6          | 26.6          |
| $V_{L,1}$ (5 V)     | 0.399      | 0.399     | 0.407    | 2.1           | 2.1           |
| $V_{L,2}$ (3 V)     | 0.511      | 0.511     | -0.555   | 208.6         | 208.6         |
| $V_{L,1} + V_{L,2}$ | 0.910      | 0.910     | -0.148   | 116.3         | 116.3         |

Table 4: Table 4.3 — Task 3, superposition (diode, Figure 4.6), in volts. The “Simulated” both-sources value is the flipped-diode .op result; single-source values are model estimates.

| Parameter           | Simulated (est.) | Measured | %diff (S v M) |
|---------------------|------------------|----------|---------------|
| $V_L$               | 0.767            | 0.233    | 69.6          |
| $V_{L,1}$ (5 V)     | 0.268            | 1.260    | 370.1         |
| $V_{L,2}$ (3 V)     | 0.376            | −2.225   | 691.8         |
| $V_{L,1} + V_{L,2}$ | 0.644            | −0.965   | 249.8         |

## Observations & Discussion

**Thévenin equivalent (Task 1).** Task 1 came out well. Using the 1 k $\Omega$  and 4.7 k $\Omega$  readings I got  $V_{th} = 1.93$  V and  $R_{th} = 2.29$  k $\Omega$ , which predicts 0.946 V across a 2.2 k $\Omega$  load. The measured value was 0.954 V, so the prediction was off by less than 1%. That is good evidence that two load measurements are enough to characterize a linear source. The Thévenin voltage came out a bit higher than the 1.78 V I got from nominal values, probably because the bench supply sat a little above 3 V and the resistors are only 5% parts.

**Linear superposition (Task 2).** For the linear circuit, the 5 V-only reading ( $V_{L,1} = 0.407$  V) was close to the calculated 0.399 V, within about 2%. The 3 V-only reading matched in size (0.555 V vs. 0.511 V) but I recorded it as negative. Since both sources push node  $Q$  above node  $P$ , the two parts should have the same sign, so the negative is most likely the voltmeter leads being reversed on that reading. The both-sources value I measured (0.668 V) came out lower than the calculated 0.910 V, and lower than the sum of the two single-source readings (0.962 V once the sign is fixed). Even with those issues, the single-source numbers line up with theory, so superposition does hold for this circuit. The gaps look like measurement error rather than a problem with the principle.

**Nonlinear superposition (Task 3).** Once the diode is in the circuit, superposition no longer holds. The sum of the single-source measurements (−0.965 V) is nothing like the both-sources reading (0.233 V). That is expected, because a diode is nonlinear: its roughly 0.7 V turn-on voltage does not scale with the sources, so the two single-source cases cannot just be added back together. The simulation shows the same thing more gently: the conducting diode gives 0.767 V across the load with both sources versus 0.644 V for the sum of the single cases.

One thing worth flagging about Task 3 is the diode direction. The simulated diode had to be turned around from the way it was first drawn, and it is easy to put a diode in backwards on a breadboard, so this is a likely source of the large simulated-versus-measured differences. It is possible the original schematic orientation was the correct one. In that case the diode stays reverse-biased and the load voltage sits near zero in all three cases, so the Task 3 mismatch would mostly be an orientation or placement problem between the simulated and built circuits rather than anything about superposition itself.

**Which circuit does superposition hold for?** It holds for the linear circuit (Figure 4.5) and fails for the diode circuit (Figure 4.6). What matters is whether the circuit is linear.

**Sources of error.** The main ones are the 5% resistor tolerances, the bench supply being slightly off from its nominal value, and voltmeter lead polarity and loading. For the diode circuit, the orientation and placement of the diode and the sensitivity of its operating point to the exact model and temperature add to this.